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Competition between Hydrogen and Halogen Bonding Interactions: Theoretical and Crystallographic Studies

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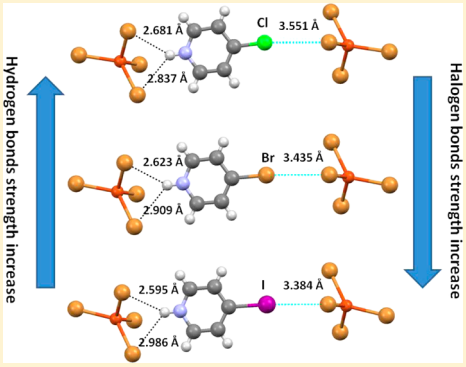
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Supporting Information

ABSTRACT: Crystal structures of six iodopyridinium tetrahalocuprate(II) salts are reported, $(n\mathrm{IP})_2\mathrm{CuX}_4$, where X = Cl or Br, *n*IP is the *n*-iodopyridinium cation, and *n* = 2, 3, or 4. The supramolecular structure of these salts is developed based on N−H···X hydrogen bonding and C−I···X halogen bonding interactions. Comparing these structures with the previously published structures of the general formulas $(n\mathrm{CP})_2\mathrm{CuX}_4$ and $(n\mathrm{BP})_2\mathrm{CuX}_4$, where $n\mathrm{CP}^+$ and $n\mathrm{BP}^+$ are the *n*-chloropyridinium and *n*-bromopyridinium cations, respectively, allows us to investigate the competition between the halogen and hydrogen bonding interactions. Henceforth, the general formula $(n\mathrm{YP})_2\mathrm{CuX}_4$ will be used to represent the 18 structures where $n\mathrm{YP}^+$ is the *n*-halopyridinium cation. Isomorphism has been observed in these structures. Isomorphic structures are divided into four sets. Analysis of the isomorphic structures allows us to apply the separation of variables principle; upon comparison of isomorphic structures, complications arise from geometrical factors due to the isomeric nature of the $n\mathrm{YP}^+$ cation and effects of intermolecular forces other than N−H···X hydrogen bonding, and C−I···X halogen bonding interactions are minimized and hence can be ignored. Comparing halogen and hydrogen bonding interaction parameters within each isomorphous set allows us to investigate the competition between these interactions. As the organic halogen becomes heavier and the halide ligand is unvaried, the N···X distance is either unvaried or becomes longer. In contrast, the Y···X distance becomes shorter even though heavier halogens have a larger radius. For example, for the isomorphous structures $(2\mathrm{BP})_2\mathrm{CuCl}_4$ and $(2\mathrm{IP})_2\mathrm{CuCl}_4$, the N···Cl distances are 2.926 Å and 3.070 Å, respectively, whereas the corresponding Y···Cl distances are 3.322 Å and 3.316 Å. Theoretical calculations have shown that bifurcated hydrogen bonding interactions are stronger than the corresponding linear ones. Also, calculations have shown that as the organic halogen becomes heavier, the halogen bonding interactions become stronger. This agrees with crystal structure data; as the organic halogen gets heavier and the halide ligand is unvaried, the difference between the two legs of the bifurcated hydrogen bond becomes larger (weaker hydrogen bonding interactions). For example, the three $(4\mathrm{YP})_2\mathrm{CuBr}_4$ structures are isomorphous; the difference between the two legs of the hydrogen bond are 0.117 Å, 0.191 Å, and 0.246 Å for $(4\mathrm{CP})_2\mathrm{CuBr}_4$, $(4\mathrm{BP})_2\mathrm{CuBr}_4$, $(4\mathrm{IP})_2\mathrm{CuBr}_4$, respectively. Surprisingly, the above two trends are valid in all isomorphous sets without exception, which is rare in solid state chemistry. Analysis of the Cu−X bond distances indicates that the Cu−X bond distance of the halogen acceptor is always shorter than that of the corresponding proton acceptor; which agrees with the theoretical calculations; hydrogen bonding interactions are stronger than the corresponding halogen bonding interactions.



INTRODUCTION

Noncovalent interactions are the core of supramolecular chemistry.[1] These intermolecular forces include hydrogen bonding, halogen bonding, $\pi-\pi$ stacking, dipole···dipole interactions, and many other such forces. Self-assembly, and hence the supramolecular structure, has received much attention in recent years;[2] it is clear that the physical properties of solid state materials are not only controlled by the identity of the structural unit but also by the spatial arrangement of those units. For example, the bulk magnetic and electrical properties of condensed matter are determined predominately by the supramolecular structure rather than the molecular structure alone.[3,4] Another intriguing phenomenon observed in solid state materials related to the intermolecular forces is polymorphism, that is, different arrangements of the same crystalline species within the crystalline lattice. Different polymorphs have different physical properties such as solubility, melting point, etc. This phenomenon has attracted extensive interest in the pharmaceutical industry, since the biological activity of pharmaceutical products is determined by supramolecular structure as well as molecular structure.[5]

Halogen bonding is a noncovalent interaction between a covalently bound halogen atom (halogen donor) and an atom with negative character (halogen acceptor); the interaction is

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Table 1. Summary of Data Collection and Refinement Parameters

crystal	2IP-Cl	3IP-Cl	4IP-Cl	2IP-Br	3IP-Br	4IP-Br
formula	$C_{10}H_{10}Cl_4CuI_2N_2$	$C_{10}H_{10}Cl_4CuI_2N_2$	$C_{10}H_{10}Cl_4CuI_2N_2$	$C_{10}H_{10}Br_4CuI_2N_2$	$C_{10}H_{10}Br_4CuI_2N2$	$C_{10}H_{10}Br_4CuI_2N_2$
form. wt	617.34	617.34	617.34	795.18	795.18	795.18
D_{calc} (Mg/m^3)	2.290	2.348	2.271	2.761	2.810	2.753
T (K)	293	293	293	293	293(2)	293(2)
crystal syst	monoclinic	triclinic	orthorhombic	monoclinic	triclinic	monoclinic
space group	$C2/c$	$P\bar{1}$	$Fdd2$	$C2/c$	$P\bar{1}$	$C2/c$
a (Å)	13.4594(6)	7.7964(6)	14.6665(11)	14.1038(19)	7.9453(7)	16.7838(10)
b (Å)	10.0648(5)	7.8950(6)	28.892(4)	10.0821(14)	8.1396(6)	8.0056(5)
c (Å)	13.2260(7)	16.4044(12)	8.5210(7)	13.462(2)	16.7670(15)	14.4175(11)
α (°)	90	90.383(6)	90	90	84.279(7)	90
β (°)	92.126(4)	95.640(6)	90	92.078(14)	83.506(7)	97.994(5)
γ (°)	90	119.440(8)	90	90	60.851(8)	90
V (Å^3)	1790.44(15)	873.33(11)	3610.8(6)	1912.9(5)	939.70(14)	1918.4(2)
ind refl	1561	3079	1331	1675	3308	1688
R(int)	0.0235	0.0199	0.0237	0.0328	0.0337	0.0517
Z	4	2	8	4	2	4
goodness of fit	1.044	1.075	1.054	1.000	1.016	1.051
$R_1{}^a$ [$I > 2\sigma$]	0.0248	0.0419	0.0294	0.0337	0.0505	0.0581
$wR_2{}^b$ [$I > 2\sigma$]	0.0521	0.0929	0.0679	0.0723	0.1115	0.1402
μ mm^{-1}	5.253	5.385	5.209	12.714	12.940	12.677
largest diff peak and hole (e·Å^{-3})	0.354 and −0.514	3.232 and −2.492	1.166 and −0.692	0.975 and −1.003	3.503 and −2.701	1.684 and −1.539

$^aR_1 = \sum||F_o| - |F_c||/\sum|F_o|$. $^bwR_2 = \{\sum[w(F_o^2 - F_c^2)^2]/\sum[w(F_o^2)^2]\}^{1/2}$.

represented as D−Y···A (Y = Cl, Br, and I) where D−Y and A are the halogen bond donor and acceptor, respectively. The halogen acceptors span a wide spectrum of species.[6,7] This includes carbonyls, tetrafluoroborates, π-systems, halide ions, halogen atoms, halide ligands, cyanide ligands, etc.[8−20] Halogen bonding interactions have been utilized in synthesis of organic conductive electrical materials,[4,21,22] topological chemistry,[23] and layer by layer assembly and chemical separation.[24,25] Also, its role in biological molecules and as a potential tool in drug design has been investigated.[26,27] Recently, halogen bonding has been exploited in nanoscience.[19,28,29] In this context, Shirman et al. utilized halogen bonding to self-assemble metal nanoparticles.[29]

Halogen bonding interactions are comparable to hydrogen bonding interactions in their strength and role in self-assembly of solid state materials.[1,10,30,31] Hydrogen bonding interaction strengths vary from weak (4 kJ/mol) to very strong (120 kJ/mol) depending on the nature of proton donor and acceptor.[1] Similarly, the calculated halogen bonding interaction energies range from weak (∼3 kJ/mol) for C−Cl···Cl−C interactions to very strong (154 kJ/mol) for C−I···F$^-$ interactions.[10,30] The thermodynamic parameters of some halogen bond interactions have been determined experimentally,[32,33] for example, the enthalpy and entropy of formation of certain types of C−I···F−M (M = Ni, Pd, and Pt; $\Delta H°$ between −25 and −16 kJ mol^{-1} and $\Delta S°$ between −73 and −49 J K^{-1} mol^{-1}).[32,33] Similarly, the C−Y···X interactions were found to compete and complement the role of the strong classical N−H···X hydrogen bonding interactions. Even theoretical calculations and analysis of (nYP) [where nYP = n-halopyrdiniumcation and n = 2, 3, or 4] crystal structures revealed that C−Y···X is a better crystal engineering tool than the corresponding N−H···X; the C−Y···X angle is closer to the linear arrangement than the corresponding N-H···X.[34]

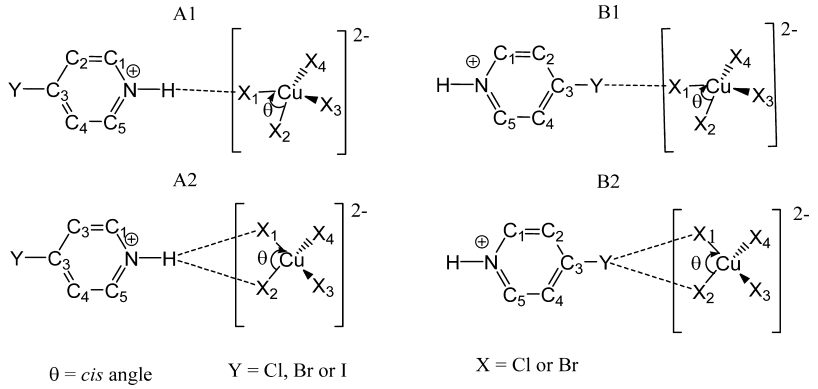
Our research group, as well as others, have focused on a special type of halogen bonding, C−Y···X−M where Y = Cl, Br, I; X = halide ligand; M = metal cation.[8,9,11−14,35−38] The characteristic feature of these interactions is that the angle C−Y···X is close to linear and the Y···X distance is less than the sum of van der Waals radii. The Y···X−M angle has preferred values in certain systems,[8,37] whereas in other systems it does not.[39] These interactions play a crucial role in the supramolecular assembly of hybrid organic−inorganic materials. This linear arrangement can be rationalized using calculated electrostatic potentials; theoretical calculations showed the presence a positive electrostatic potential end cap on the carbon-bonded halogen atom; in contrast, the electrostatic potential is negative on the π-region of the halogen atom.[11,30,40] Hence, the linear arrangement is associated with attractive electrostatic forces.

The competition between halogen bonding and hydrogen bond interactions has been studied experimentally and theoretically.[8,31,41−45] Aakeröy et al. have investigated the competition between halogen and hydrogen bonding interactions in molecular cocrystals of organic compounds by varying the halogen bond donor and keeping the hydrogen bond donor and acceptors (hydrogen and halogen bond) unvaried. Their results indicated that strong halogen bonding interactions C−I···N prevail over C−H···N interactions, whereas the weaker C−Br···N halogen bonding interactions do not.[45] In a more recent study, based on cocrystals of 3,3′-azobipyridine and 4,4′-azobipyridine with bifunctional halogen and hydrogen bonding donors, they have shown that the binding-site location plays a significant role in the overall balance between halogen and hydrogen bonding interactions.[41] Also, Voth et al. studied hydrogen and halogen bonding interactions in biomolecules, in the same molecular environment; halogen bonding interactions are more stable by 2−5 kcal/mol than the analogous hydrogen bonding interactions.[42] Brammer et al. and Awwadi et al. studied the competition between halide and halometallate ions as halogen and hydrogen bond acceptors in mixed anion inorganic materials.[8,38,43,46,47] Their research indicated that the halide ion prevails over the halometallate ion as a hydrogen bond acceptor.

We have investigated the role of C−Y···X−Cu interactions in the control of the crystal structures of (nYP)$_2$CuX$_4$ Y = Cl, Br;

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Scheme 1. Parameters Used for Modeling (A) Hydrogen Bonding and (B) Halogen Bonding Interactions



X = Cl, Br.[11,39] In this report, complementing our previous work, the crystal structures of six bis(*n*-iodopyridinium)-tetrahalocuprate salts, $(n\text{IP})_2CuX_4$, are investigated [(*n*IP = *n*-iododpyridinium cations, *n* = 2, 3, and 4): $(2\text{IP})_2CuCl_4$, henceforth, 2IP-Cl; $(2\text{IP})_2CuBr_4$, henceforth, 2IP-Br; $(3\text{IP})_2CuCl_4$, henceforth, 3IP-Cl; $(3\text{IP})_2CuBr_4$, henceforth, 3IP-Br; $(4\text{IP})_2CuBr_4$, henceforth, 4IP-Br; and $(4\text{IP})_2CuCl_4$, henceforth, 4IP-Cl]. The crystal structures are reported in this paper and, along with related compounds published previously, are categorized into isomorphic sets. This enabled our investigation of the competition between halogen and hydrogen bonding interactions. Our discussion will be supported by theoretical calculations.

EXPERIMENTAL SECTION

Synthesis and Crystal Growth. *2IP-Cl, 2IP-Br, 3IP-Br, 4IP-Cl, and 4IP-Br.* A general procedure has been followed to prepare these five salts. One mmole of copper halide ($CuBr_2$ or $CuCl_2{\cdot}2H_2O$) was dissolved in 15 mL of methanol. This solution was acidified with 1 mL of the corresponding concentrated hydrohalic acid. Another solution was prepared by dissolving 2 mmol of the corresponding iodopyridine in 15 mL of methanol, which was then acidified with 1 mL of the corresponding hydrohalic acid. The two solutions were mixed and stirred while being gently heated for 10 min. This mixture was filtered and left to evaporate slowly. After a few days, crystals formed, which were collected by filtration, washed with methanol, and allowed to air-dry. A suitable crystal of each was selected for collecting X-ray diffraction data. The yields were in the range of 40−60%.

3IP-Cl. One mmole (0.13 g) of $CuCl_2$ was dissolved in 10 mL of acetonitrile and then acidified with 1 mL of concentrated hydrochloric acid. Two mmoles (0.41 g) of 3-iodopyridine were dissolved in 10 mL of acetonitrile. The two solutions were mixed and heated for 15 min and then filtered while hot. The solution was left to cool at room temperature; a few yellow crystals formed within 30 min. The filtrate was left to evaporate slowly forming additional yellow parallelepiped crystals along with sky-blue wool-like crystals. Crystal structure analysis indicated that the yellow crystals are 3IP-Cl. The blue crystals are $(3\text{IP})_2[CuCl_3(H_2O)_2]Cl$.[48]

Crystal Structure Determination. The diffraction data sets of the six salts were collected at room temperature using an Oxford Xcalibur diffractometer (Mo Kα radiation, λ = 0.7107 Å). Data were acquired and processed to give *hkl* files using CrysAlisPro software.[49] The structures were solved by direct methods and refined by the least-squares method on F^2 using the SHELXTL program package.[50] Hydrogen atoms were placed in calculated positions and refined isotropically using a riding model. All non-hydrogen atoms were refined anisotropically. Details of the data collection and refinement are given in Table 1.

Theoretical Method. Gaussian 03 was used for geometry optimization and single point calculations.[51] The molecular units of the cations [$4YP^+$] and anions [CuX_4^{2-}] were optimized at the DFT/B3LYP level with the cc-pvdz basis set on C, N, Cl, and H and the lanl2dz basis set on, Br, I, and Cu, where $4YP^+$ = 4-halopyridinium cation, Y = Cl, Br, or I and X = Cl or Br. Energies of interactions were calculated for both linear and symmetrical bifurcated charge assisted hydrogen and halogen bonds (Scheme 1). The calculated energy values include both energy due to direct cation−anion electrostatic forces and those due to other intermolecular forces which are mainly either hydrogen or halogen bonding interactions. For simplicity, these charge assisted halogen and hydrogen bonding interactions are named halogen and hydrogen bond interactions. The energy of interaction was calculated as function of the inter-cation−anion distances and the torsion angles X3−Cu···N−C1, X3−Cu···C3−C2, X1−Cu···N−C1, and X1−Cu···C3−C2 for $4CP{\cdots}CuX_4^{2-}$ systems (Scheme 1). The results (vide infra) indicated that effect of the torsion angle is negligible in the linear systems; in contrast, the torsion angles play a significant role in the bifurcated systems. Hence, the torsion angles X3−Cu···N−C1 of the linear hydrogen bonding interactions and X3−Cu···C3−C2 of the linear halogen bonding interactions were constrained to 90° (Scheme 1A1,B1). Also, the energy of the charge assisted bifurcated pattern was calculated for two different arrangements: (a) the torsion angles X1−Cu···N−C1 (Scheme 1A2) and X1−Cu···C3−C2 (Scheme 1B2) are constrained to 0°; henceforth, the arrangement is the planar bifurcated pattern. (b) The torsion angles X1−Cu···N−C1 (Scheme 1A2) and X1−Cu···C3−C2 (Scheme 1B1) are constrained to 90° and henceforth is the perpendicular bifurcated pattern. The energy of interaction for both linear hydrogen bonding and linear halogen interaction was calculated using the formula

$$E_{int} = E_{CatAn} - E_{Cat} - E_{An}$$

where E_{CatAn} is the total energy of the interacting $4YP^+$ cation and CuX_4^{2-} anion, and E_{Cat} and E_{An} are the energies of separate cations and anions, respectively.

RESULTS

Crystallographic Results. The six crystal structures consist of tetrahalocuparate(II) anions and iodopyridinium cations. The tetrahalocuprate anions show perfect *D2d* geometry except in 3IP-X, where it shows approximate *D2d* symmetry. The trans angles average 137.1° (range 131.6−141.3°). The trans angles in the chloride salts are larger than those in the corresponding bromide salts; the averages are 139.1° and 135.1° for the chloride and bromide salts, respectively. The Cambridge Structural Database CSD version 5.35 update (November 2013) was searched for the trans angles in $CuCl_4^{2-}$ and $CuBr_4^{2-}$ structures. The histogram distribution of number of hits versus trans angle (α) range is shown in Figure 1. For $CuCl_4^{2-}$ and $CuBr_4^{2-}$, the maximum number of hits occurs between 130 and 134.9°. However, the data suggest that the trans angles in $CuCl_4^{2-}$ are wider than in $CuBr_4^{2-}$ by less than 5° since, in $CuCl_4^{2-}$ structures, the number of hits between

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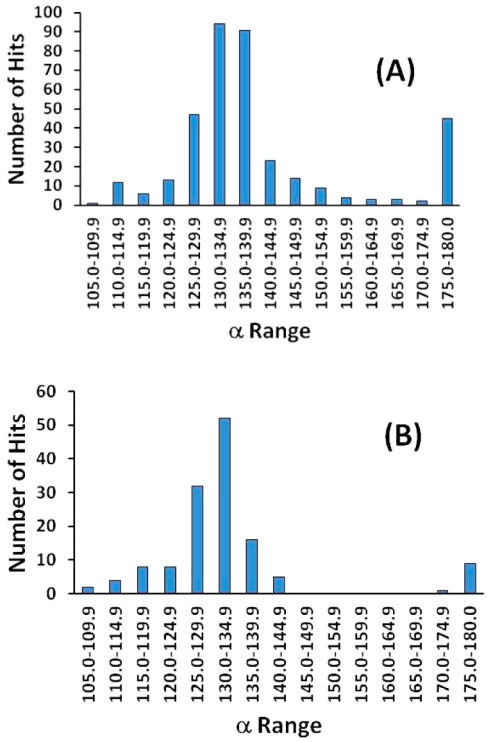


Figure 1. Histogram distribution for the number of hits versus average X−Cu−X trans angles (α) for all reported CuX_4^{2-} structures in Cambridge Structural Database: (A) $CuCl_4^{2-}$ and (B) $CuBr_4^{2-}$. If a structure was determined more than once, only one of these structures was involved in the statistical analysis. It is noteworthy that this statistical analysis using an older version of CSD for $CuCl_4^{2-}$ structures was published in a previous work; we include it here for sake of comparison.[52]

135.0 and 139.9° is higher than that between 125.0 and 129.9° (Figure 1A). In contrast, the number of hits between 135.0 and 139.9° in $CuBr_4^{2-}$ structures is smaller than that between 125.0 and 129.9° (Figure 1B).[52] The two cations are crystallogaphicaly equivalent in all analyzed structures except in 3IP-X where the two cations are crystallographically independent and henceforth are labeled as cat1 and cat2. The 2IP-X salts are isomorphous, and therefore only one of them will be used to illustrate the supramolecular structure. The structures of the two 3IP-X salts are not isomorphous, even though the cell parameters are similar (Table 1). The differences in the cell angles of the two 3IP-X salts can be minimized by transforming the cell angles of one the structures to a nonstandard setting. The unit cell parameters of 3IP-Br can be transformed to 7.9453(7) Å, 8.1396(6) Å, 16.7670(15) Å, 84.279(7)°, 96.494(7)°, 119.149(8)°. However, even though they are not isomorphous, their supramolecular structures are similar, and again only one will be used to illustrate the structures.

The self-assembly of these salts is rationalized using mainly N−H···X−M hydrogen bonding and C−I···X−M halogen bonding interactions (Figure 2). The data summarizing these interactions are listed in Table 3. There are two patterns of N−H···X−M hydrogen bonding interactions, linear and bifurcated. The linear pattern is observed in the 2IP-X and 4IP-Cl structures (Figure 2A,C), while the bifurcated pattern is observed in 3IP-X and 4IP-Br (Figure 2B,D). C−I···X^- interactions are characterized by essentially linear C−I···X^- angles (avg = 167.9°; range 152.8−176.9°), and I···X^- distances are 0.18−0.52 Å less than the sum of van der Waals radii (I = 1.98 Å; Br = 1.85 Å; Cl = 1.75 Å). In the 3IP-X structures, although the two cations are involved in bifurcated hydrogen bonding, the pattern is asymmetrical bifurcated in cat1, whereas it is symmetrical in cat2 as indicated by N···X distances (Table 3).

Iodopyridinium cations serve as bifunctional supramolecular synthons (excluding the weaker stacking interactions and C−H···X hydrogen bond) via two intermolecular interactions, the N−H···X and C−I···X interactions. Stacking interactions include N(π)···X and π···π interactions. Using these supramolecular synthons, all reported structures form well-defined extended structures. These extended structures accumulate into the final crystal structure via those stacking interactions as well as other weaker intermolecular interactions such as C−H···X hydrogen bonds.

The observed extended structures in the studied salts can be categorized into three patterns: (a) extended chains; this pattern is observed in 2IP-Cl, 2IP-Br, and 4IPB-Br (Figure 3). The connectivity in these three chain structures are similar with the exception that the hydrogen bonding pattern is bifurcated in 4IPB-B (Figure 3B), whereas it is linear in the 2IP-X salts (Figure 3A). Because of the higher symmetry of the 4IP cation, the 4IP-Br chain structure is more compact (Figure 3B). In the 2IP-X structures, the chains run parallel to the *c*-axis, while in 4IP-Br the chains run parallel to −1 0 1 direction. (b) Ladder structure patterns observed in 3IP-X (Figure 4); the rungs are made by cat 1 and the rails by cat2. Strong bifurcated hydrogen bonding interactions are involved in the rails, whereas strong halogen bonding interactions are involved in the rungs. The ladders lie approximately parallel to the 1 2 4 plane in 3IP-Br, whereas they lie approximately parallel to the −2 −2 7 plane in 3IP-Cl. (c) The network structure in 4IP-Cl is the most complicated. The best way to visualize this structure is to describe it as extended helices which twist counter clockwise (left-handed) as shown in Figure 5A. The helical structure extends parallel to the *c*-axis (Figure 5B). These helices are linked by both halogen and hydrogen bonding interactions to form a three-dimensional extended structure (Figure 6).

Two additional intermolecular interactions connect the extended structures to form the three-dimensional structures: nonclassical C−H···X^- hydrogen bonding interactions and π-

Table 2. Selected Bond Distances and Angles

crystal	2IP-Cl	3IP-Cl	4IP-Cl	2IP-Br	3IP-Br	4IP-Br
trans angle 1 (°)	134.16(4)	141.28(10)	141.02(9)	133.92(3)	139.10(8)	131.58(5)
trans angle 2 (°)	134.16(4)	143.11(8)	141.02(9)	133.92(3)	140.28(7)	131.58(5)
Cu−X1 distance (Å)	2.262(1)	2.266(2)	2.250(2)	2.389(1)	2.355(2)	2.384(2)
Cu−X2 distance (Å)	2.241(1)	2.255(2)	2.241(2)	2.376(1)	2.396(2)	2.359(1)
Cu−X3 distance (Å)		2.273(2)			2.397(2)	
Cu−X4 distance (Å)		2.221(2)			2.391(2)	

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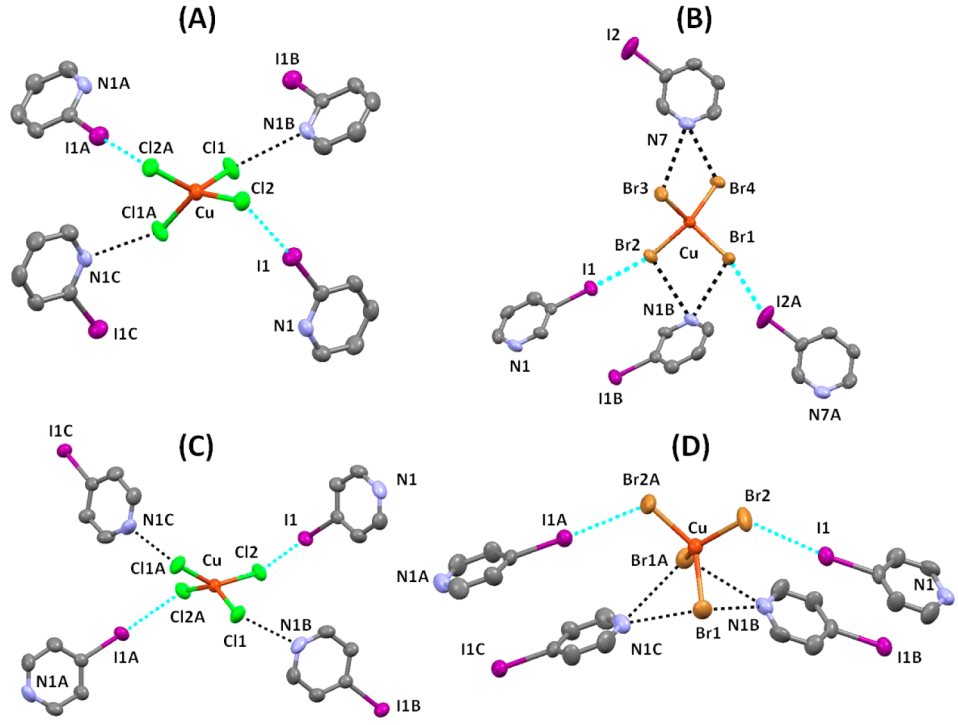


Figure 2. Synthon interactions in (A) 2IP-Cl, (B) 3IP-Br, (C) 4IP-Cl, and (D) 4IP-Br. Hydrogen and halogen bonding interactions are represented by black and blue dotted lines, respectively. Thermal ellipsoids are shown at 50% probability.

Table 3. C−I···X and N−H···X Synthons Distances (Å) and Angles (°) in the *n*IP$^+$ Salts

compound	I···X	C−I···X	I···X−Cu	N···X	H···X	N−H···X	H-bond type
2IP-Cl	3.316(1)	176.88(10)	93.47(3)	3.070(3)	2.216	172.14	linear
3IP-Cl···cat1	3.237(2)	168.73(19)	161.20(8)	3.089(1)	2.323	148.53	bifurcated
				3.408(6)	2.968	113.83	
3IP-Cl···cat2	3.549(2)	155.73(20)	151.16(12)	3.262(6)	2.680	126.04	bifurcated
				3.263(6)	2.477	152.23	
4IP-Cl	3.318(2)	176.12(18)	106.10(7)	3.072(7)	2.232	165.34	linear
2IP-Br	3.414(1)	175.63(0.18)	93.47(3)	3.212(5)	2.357	172.28	linear
3IP-Br cat1	3.364(1)	169.50(26)	163.64(5)	3.255(8)	2.490	148.69	bifurcated
				3.546(9)	3.068	117.31	
cat2	3.635(2)	152.79(30)	146.18(7)	3.433(9)	2.641	153.72	bifurcated
				3.435(9)	2.869	124.92	
4IP-Br	3.384(1)	174.69(30)	111.73(5)	3.367(9)	2.595	149.99	bifurcated
				3.613(10)	2.986	131.45	

stacking interactions. The role of the C−H···X$^-$hydrogen bonds will not be discussed due to the abundance of hydrogen atoms in the iodopyridinium cations; this interaction is expected to be present in all of these structures. Three parameters are defined to analyze the N(π)···X and π···π interactions. In the case of N(π)···X interactions, these parameters are the nitrogen to halide distance (dN−X), the perpendicular distance between the halide ligand and the plane of the pyridinium cations, and the angle between these two vectors, γ. For π−π interactions, the three parameters are defined in a similar manner, except that the centroid to centroid (dc−c) distance is used rather than the dN−X distance.[11] These interactions are shown in Figures 7 and 8, and the data summarizing them are listed in Table 4. γ and Φ values are avg = 21.2° range = 12.0−29.2° and avg = 23.9 °, range = 11.6 −39°, respectively. N(π)···X interactions are present in all structures except 3IP-Br. In contrast, π···π stacking is minimal or absent in all structures except in 3IP-X (Figure 8).

Theoretical Results. The optimized molecular geometry of the tetrahalocuprate(II) anion is flattened tetrahedral with *D2d* symmetry; the cis and trans angles are 100.8° and 128.8° for [$CuCl_4^{2-}$] and 101.2° and 127.6° for [$CuBr_4^{2-}$]. Also, the fact that the trans angles in the [$CuCl_4^{2-}$] anion are larger than that in the [$CuBr_4^{2-}$] agrees with experimental values (vide supra). The calculated energies of linear interactions have shown the effect of the torsion angle X3−Cu···N−C1 and X3−Cu···C3−C2 on the calculated energies values is negligible (Scheme 1 and Figure 9); hence, for other systems (4BP−CuX_4^{2-} and 4IP−CuX_4^{2-} systems), these angles are constrained to 90° (Figure 10A1,B1). The energies of interaction are listed in Table 5. The torsion angles X1−Cu···N−C1 and X1−Cu··· C3−C2 play a significant role in bifurcated systems (Figure 8); for hydrogen bonding interactions, the planar bifurcated interactions (Figure 10A2) are more stable than the corresponding perpendicular bifurcated interactions (Figure 10A3) by 18 kJ/mol in 4CP$^+$−$CuCl_4^{2-}$ and 14 kJ/mol in

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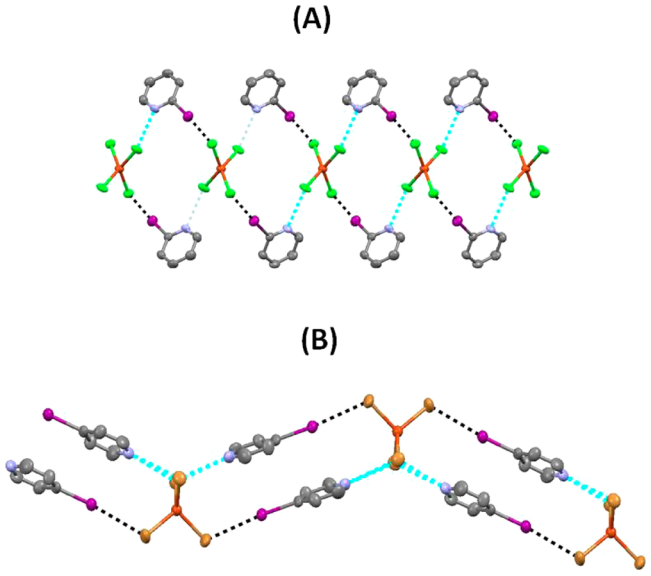


Figure 3. The extended structures in (A) 2IP-Cl and (B) 4IP-Br. N–H···X and C–I···X interactions are represented by blue and black dashed lines.

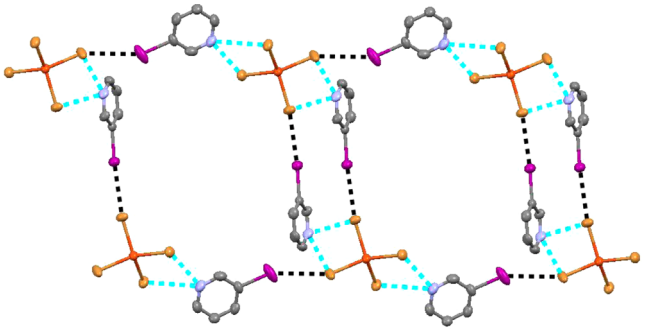


Figure 4. The ladder structure in 3IP-Br. N–H···X and C–I···X interactions are represented by blue and black dotted lines.

$4CP^{+}$–$CuBr_4^{2-}$ (Figure 9A). In contrast, the perpendicular bifurcated halogen interactions (Figure 10B3) are more stable than the corresponding bifurcated ones (Figure 10B2) by 19 kJ/mol in $4CP^{+}{\cdot}CuCl_4^{2-}$ and 17 kJ/mol in $4CP^{+}{\cdot}CuBr_4^{2-}$ (Figure 9B). Then, the energies of interactions were calculated at two different arrangements in other systems ($4BP{\cdot}CuX_4^{2-}$ and $4IP{\cdot}CuX_4^{2-}$ systems), planar bifurcated (Figure 10A2,B2) and the corresponding perpendicular bifurcated interactions (Figure 10A3,B3). The calculated energies are listed in Tables 5 and 6.

Hydrogen bond interactions are always more stable than the corresponding halogen bonding interaction by 105–197 kJ/mol. This can be rationalized based on the fact that the positive charge is more concentrated on the N–H group. This is supported by the calculated electrostatic potential surface of 4-bromopyridinium and 4-chloropyridinium cations; the highest electrostatic potential values are around the N–H group.[36] As expected, for both hydrogen and halogen bond interactions the energy of the interaction is stronger in the $[CuCl_4^{2-}]$ anion than in the corresponding $[CuBr_4^{2-}]$ anion. This because $[CuCl_4^{2-}]$ is smaller in size, and hence the negative charge is more concentrated.

The bifurcated interactions are always more stable than the corresponding linear pattern. In bifurcated patterns, the cations

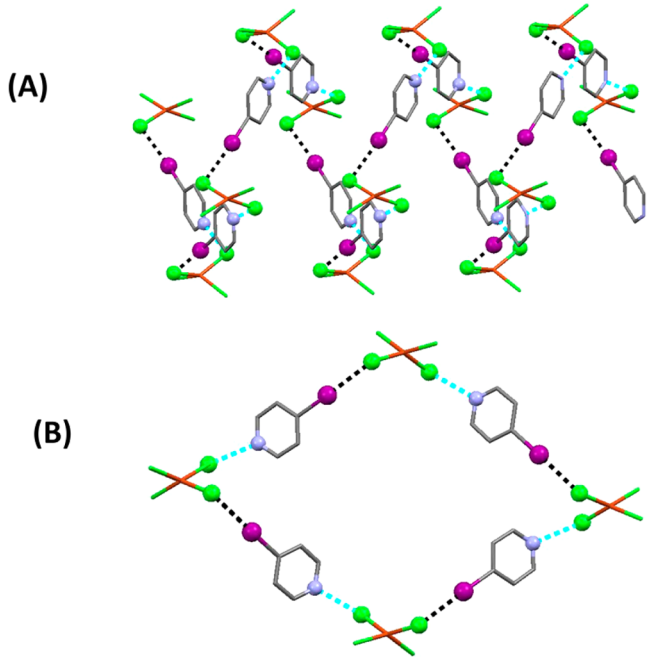


Figure 5. Different views of the helical structures in 4IP-Cl; (A) side view and (B) along the helical structure. The helices extend parallel to the *c*-axis.

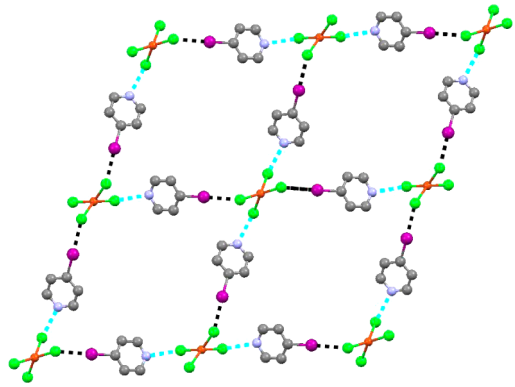


Figure 6. The extended structure of 4IP-Cl viewed along the *c*-axis.

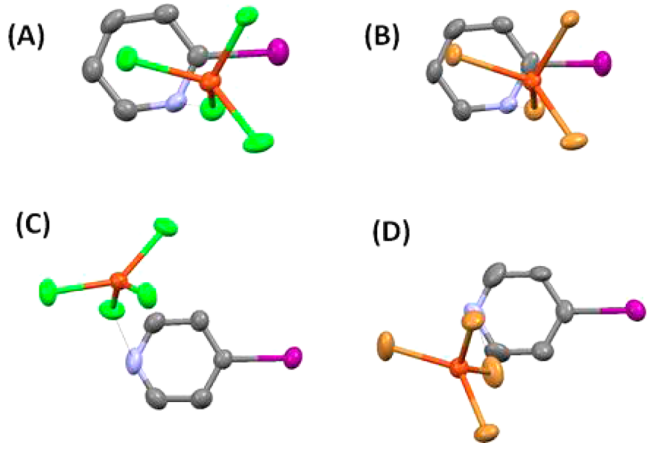


Figure 7. (A–D) Illustrations of N(π)···X interactions. Views are perpendicular to the plane of the cation.

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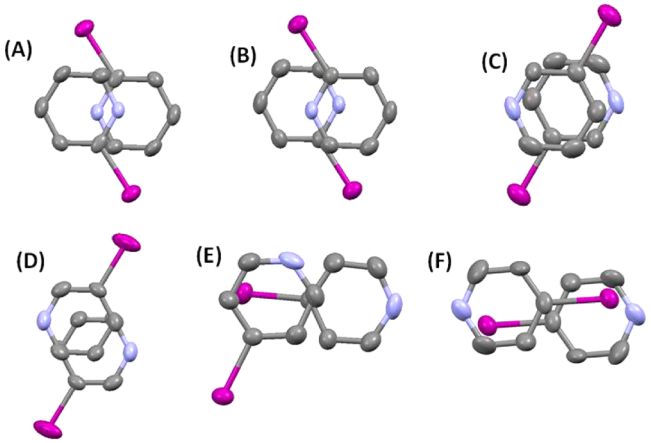


Figure 8. Illustration of $\pi-\pi$ stacking interactions; (A) 2IP-Cl, (B) 2IP-Br, (C) 3IP-Br-cat1, (D) 3IP-Br-cat2, (E) 4IP-Cl, and (F) 4IP-Br. Views are perpendicular to the plane of the cation.

and anions approach each other more closely and hence generate more stable interactions. On average, the pla.bifurcated and the perpendicular bifurcated interactions are more stable than the corresponding linear interactions by 52 kJ/mol. This energy difference is more pronounced in the hydrogen bonding interactions; the average energy difference between linear and bifurcated hydrogen bonding is 82 kJ/mol, whereas the average difference in the case of halogen bonding is 22 kJ/mol. The planar bifurcated hydrogen bonding pattern is more stable than the corresponding perpendicular one by ca. 16 kJ/mol; in contrast, perpendicular pattern halogen bonding interactions are more stable than the bifurcated ones by ca. 18 kJ/mol. The extra stability of the planar bifurcated hydrogen bonding interactions over the corresponding perpendicular ones may be attributed to the presence of C1−H···X1 and C5−H···X2 nonclassical hydrogen bonding interactions (Figure 10A2,A3). The effect of the analogous nonclassical hydrogen bonding interactions (C2−H···X1 and C4−H···X1) in the halogen bonding interactions is minor (Figure 10B2,B3). This explanation is supported by the calculated electrostatic potential surface of the 4-bromopyrdinium and 4-chlorpyridinium cations; the electrostatic potential values around C−H groups adjacent to the N−H group is more positive than the other C−H groups.[36]

Energy of hydrogen bonding interactions is the same for $4CP^{+}\cdot[CuX_4^{2-}]$ and $4BP^{+}\cdot[CuX_4^{2-}]$; the differences are within 2 kJ/mol. These interactions are weaker in $4IP^{+}\cdot[CuX_4^{2-}]$ (Table 5). For example, the bifurcated hydrogen bonding interaction in $4IP^{+}\cdot[CuCl_4^{2-}]$ is weaker than that in $4CP^{+}\cdot[CuCl_4^{2-}]$ by 11.8 kJ/mol. The heavier the organic halogen

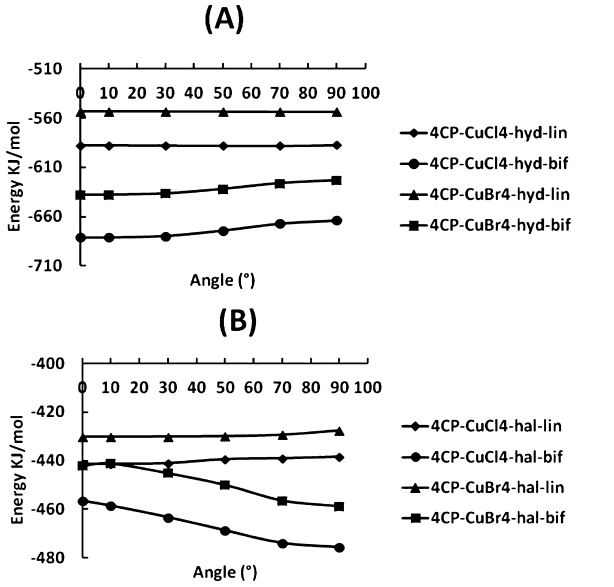


Figure 9. The calculated energy of interaction in $4CP\cdot CuX_4^{2-}$; (A) hydrogen bonding interactions, (B) halogen bonding interactions, angle = torsion angle; these torsion angles are defined in Figure 10.

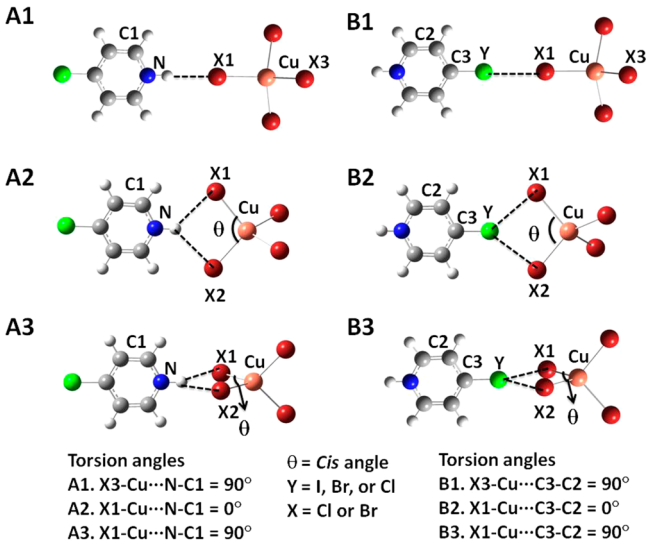


Figure 10. Geometries of the modeled (A) hydrogen bonding interactions and (B) halogen bonding interactions. A1, A2, and A3 represent the linear, planar bifurcated, and perpendicular bifurcated hydrogen bonding interactions. Similarly B1, B2, and B3 represent the linear, planar bifurcated, and perpendicular bifurcated halogen bonding interactions.

Table 4. Packing Interactions Parameters

compound	dX−N (Å)[a]	$d_\perp$	γ (°)	dc−c (Å)[b]	$d_\perp$ (Å)	Φ (°)
2IP-Cl	3.624	3.496	15.27	3.988	3.587	25.91
2IP-Br	3.604	3.524	12.01	4.087	3.608	28.02
3IP-Cl				3.589	3.478	14.29
				3.737	3.529	19.21
3IP-Br				3.603	3.530	11.55
				3.850	3.630	19.46
4IP-Cl	3.614	3.177	28.47	4.438	3.694	33.66
4IP-Br	3.826	3.339	29.22	4.765	3.703	39.00

[a]Stacking interaction is considered present, if dX−N distance is less than the sum of van der Waals radii +0.5 Å. [b]Shortest dc−c distance.

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Table 5. Calculated Energies of Hydrogen Bonding Interactions (kJ/mol)[a]

cation	4CP			4BP			4IP		
hydrogen bond	lin	pla.bif	per bif	lin	pla.bif	per bif	lin	pla.bif	per bif
$CuCl_4^{2-}$	−587.2	−681.0	−663.7	−588.3	−681.6	−664.3	−582.3	−674.9	−657.7
$CuBr_4^{2-}$	−553. 5	−637.6	−623.1	−554.1	−638.1	−623.6	−548.0	−631.7	−617.3

[a]lin = linear, pla.bif = planar bifurcated, and per bif = perpendicular bifurcated. The definition of these hydrogen bonding interactions is shown in Figure 10.

Table 6. Calculated Energies of Halogen Bonding Interactions (kJ/mol)[a]

cation	4CP			4BP			4IP		
halogen bond	lin	pla.bif	per bif	lin	pla.bif	per bif	lin	pla.bif	per bif
$CuCl_4^{2-}$	−438.32	−456.5	−475.5	−442.2	−459.8	−477.8	−461.9	−478.3	−499.9
$CuBr_4^{2-}$	−427.6	−442.0	−456.5	−432.6	−444.1	−459.4	−443.0	−454.3	−475.4

[a]lin = linear, pla.bif = planar bifurcated, and per bif = perpendicular bifurcated. The definition of these hydrogen bonding interactions is shown in Figure 9.

atom the more stable the halogen bonding interaction becomes. For example, the bifurcated halogen bonding interactions in 4IP·$[CuCl_4^{2-}]$ are stronger than those in $4CP^{+}$·$[CuCl_4^{2-}]$ by 23.4 kJ/mol (Table 6).

DISCUSSION

We have reported in this paper and previously published work 18 structures of the general formula $(n\text{YP})_2CuX_4$. Two characteristic geometric parameters are associated with C−Y···X−Cu halogen bonding interactions, the essentially linear C−Y···X angle (avg = 171.6° range 152.8−178°), and the Y···X distances are less than the sum of van der Waals radii. Y···X−M angles have preferred values in certain examples,[37] while in others they do not.[39] Contacts within the sum of van der Waals radii plus 0.1 Å were considered. Contacts with C−Y···X angles less than 90° were not included in this analysis; such contacts were only observed in 2CP−X. The averages for the C−Y···X angles are 172.8° ± 3.8°, 174.0° ± 3.6°, and 168.8° ± 9.5° for Y = Cl, Br, and I, respectively. Table 7 lists the average Y···X distances and Δ values (the difference between these distances and the sum of van der Waals radii of the corresponding atoms). Only room temperature structures are included in Table 7, since these distances are temperature dependent. The only structure that was collected at lower temperature is 3BP-Br. In all of the studied complexes, the Y···X distances are less than the sum of the van der Waals radii except in 2CP-Br and one of the two crystallographically independent cations in the $P2(1)/c$ phase of 4CP-Cl. Inspection of the data listed in Table 7 reveals that as the organic halogen atom becomes heavier, the Δ values become larger, whereas there is no obvious trend in Δ values based on the halide anion. This agrees with theoretical calculations; as Y becomes larger the halogen bonding interaction becomes stronger. Furthermore, all crystallographically independent cations are involved in a halogen bonding interaction when the organic halogen is either bromine or iodine, but in the case of a chlorine substituent, only three out of the five crystallographically independent chloropyridinium cations are involved in C−Cl···Br−Cu interactions. This is also in accordance with the observed hierarchy of these types of halogen bonding interactions.[37]

Table 7. Averages of Y···X Distances (Å) and the Difference between These Distances and the Sum of Van Der Waals Radii, Δ (Å)

compound's group	Y···X[b]	Δ[a,c]
*n*CP-Cl	3.35	0.12
*n*CP-Br	3.56	0.05
*n*BP-Cl	3.32	0.28
*n*BP-Br	3.44	0.26
*n*IP-Cl	3.36	0.37
*n*IP-Br	3.45	0.38

[a]Δ = sum of van der Waals radii -Y···X. [b]The interaction is considered if C−Y···X angle is greater than 90°. [c]All esd's in Δvalues are less than 0.005 Å.

Analysis of the $(n\text{YP})_2CuX_4$ structures along with the results of the theoretical calculations allows us to investigate the competition between halogen and hydrogen bonding interactions. Isomorphism has been observed in several structures. Therefore, to simplify the discussion of the competition between halogen and hydrogen bond interactions, the isomorphic structures within this group are classified into four isomorphous sets (Table 8). Two crystal structures are said to be isomorphous if the two structures have the same unit cell parameters, the same space group, and their atoms occupy similar coordinates. Comparing the interactions within each isomorphous structure set separately enables us to apply the separation of variables principle, which minimizes the effect of geometrical factors due to the isomeric nature of the *n*YP cations as well as the effect of other intermolecular forces.

Table 8. Sets of $(n\text{YP})_2CuX_4$ Isomorphous Structures[a]

set	space group	structures
1	$P\bar{1}$	3CP-Cl, 3BP-Cl
2	$P21/c$	4CP-Cl(I)[a], 4BP-Cl
3	$C2/c$	4CP-Cl(II)[b], 4CP-Br, 4BP-Br, 4IP-Br
4	$C2/c$	2BP-Cl, 2BP-Br, 2IP-Cl, 2IP-Br

[a]The compounds that are not isomorphous with any other structure are not included in this analysis "six compounds". [b]There are two known polymorphs of 4CP-Cl.[11]

The competition between N−H···X−M hydrogen bonding interactions and C−Y···X −M halogen bonding interactions is investigated by (a) comparing N···X and Y···X distances and (b) comparing the length of the two legs of the bifurcated hydrogen bonding interactions in each isomorphous series and then supporting the argument with theoretical results. Table 9

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Table 9. Averages of Hydrogen and Halogen Bonding Interactions Distances (Å) in *n*YP−X

compound	N···X	average N···X	β^a	Y···X⁻	hydrogen bond type	set
3CP-Cl-cat 1	3.278	3.255	0.047	3.277	bifur	1
	3.231					
Cat 2	3.149			3.341	linear	1
3BP-Cl cat 1	3.182	3.252	0.139	3.271	bifur	1
	3.321					
Cat 2	3.149			3.283	linear	1
4CP-Cl(I)	3.133	3.266	0.266	3.446	bifur	2
Cat 1	3.399					
Cat 2	3.094			3.511	linear	2
4BP-Cl	3.120	3.275	0.309	3.356	bifur	2
	3.429					
Cat 2	3.093			3.358		2
4CP-Br	3.399	3.458	0.117	3.551	bifur	3
	3.516					
4BP-Br	3.37	3.466	0.191	3.435	Bifur	3
	3.561					
4IP-Br	3.367	3.49	0.246	3.384	Bifur	3
	3.613					
2BP-Cl	2.926			3.322	linear	4
2IP-Cl	3.070			3.316	linear	4
2BP-Br	3.195			3.445	linear	4
2IP-Br	3.211			3.414	linear	4

[a]The difference between the two legs of bifurcated hydrogen bond.

lists the N···X distances and Y···X distances for all isomorphous sets.

Inspection of these data reveals two interesting observations: (a) As the organic halogen atom becomes heavier without changing the halide anion, the N···X distance becomes longer; in contrast, Y···X distances become shorter in spite of heavier halogen atoms having greater van der Waals radii. For example, Set 3 (Table 7) consists of four isomorphous structures. Three of them have the general formula 4YP-Br, Y = Cl, Br, or I. The fourth member of this set, 4CP-Cl, is excluded from this comparison since both hydrogen and halogen bonds strength are varied. The average of the N···Br distances of the two legs of the bifurcated hydrogen bond is the longest in the case of 4IP-Br, 3.490 Å (Table 9) and the shortest in the case of 4CP-Br, 3.458 Å. However, the Y···Br distance is the shortest in the case of 4IP-Br, 3.384 Å (Table 9) and the longest for 4CP-Br, 3.551 Å. This agrees with the theoretical results which indicated that as the organic halogen atoms get heavier the halogen bonding gets stronger, while the N−H···X hydrogen bonding interaction strength does not vary as much (Tables 5 and 6). The energies of hydrogen bonding interactions in 4BP-X are almost equal to those of the corresponding 4CP-X (within 2 kJ/mol), and hydrogen bonding interactions are less stable in 4IP-X by 6 kJ/mol (Table 5). In contrast, the halogen bonding interactions are more stable in 4IP-X than in the corresponding 4CP-X by 19 kJ/mol (Table 6).

(b) The bifurcated hydrogen bond deviates more from the symmetric arrangement in the case of heavier organic halogens; the difference between the lengths of the two legs of the bifurcated hydrogen bond is larger for heavier organic halogens (β value, Table 9). For example, the bifurcated hydrogen bonding is closer to symmetrical in 3CP-Cl-cat1 (difference = 0.042) than in 3BP-Cl-cat 1 (difference = 0.139). This indicates as the halogen bonding interactions get stronger, the hydrogen bonding interaction get weaker. This agrees with the theoretical calculations; bifurcated hydrogen bonding interactions are stronger than the corresponding linear interactions (Table 5).

These two trends are obeyed in all corresponding interactions in all reported isomorphous structural sets. This reflects the competition between these two interactions and supports the idea that the energies of halogen bonding interactions are comparable to hydrogen bonding interactions in these systems.

The effect of halogen bonding and hydrogen bonding interactions on the Cu−X bond distances was investigated. The Cu−X bond distances of the halogen acceptors and proton acceptors are listed in Table 10. Utilizing Cu−X bond distances as parameters to evaluate the competition between halogen bonding and hydrogen bonding interaction is not of definitive benefit, since the variation of Cu−X bond distances is not large. However, the analysis of the Cu−X bond distances indicates (a) the Cu−X bond distance in the role of the hydrogen bond acceptor is longer than the corresponding Cu−X distances of the halogen bond acceptor. This agrees with theoretical calculations; hydrogen bonding interactions are always stronger than the corresponding halogen bonding interactions. (b) In each isomorphous set, as the organic halogen becomes heavier, the Cu−X bond distance of the halogen acceptor either remains unvaried or becomes longer. In contrast, the Cu−X distance of the hydrogen bond acceptor becomes shorter with one exception (2BP-Cl and 2IP-Cl, Table 10). This agrees with previous analysis (vide supra).

Analysis of the geometric parameters for C−I···X−M halogen bonding interactions in the structures reported in this article and those published in the Cambridge Structural Database (CSD) indicates that the preferred arrangements of C−I···X−M synthons are similar to that of the well-studied C−Y···Y−C synthons.[9] There are two preferred arrangements for synthons C−Y1···Y2−C: (a) type I, C−Y1···Y2 ≅ Y1···Y2−C ≅ 150° and (b) type II, C−Y1···Y2 = 180° and Y1···Y2−C ≅ 90°. The CSD version 5.35 update (Nov 2013) has been

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Table 10. Cu−X Bond Distances (Å) for the Halogen and Proton Acceptors in Halogen and Hydrogen Bonding Interactions[a,b]

compound	halogen acceptor	proton acceptor	set
4CP-Cl(I)	2.221 ± 0.012[c]	2.264 ± 0.017[c]	2
4BP-Cl	2.228 ± 0.012[c]	2.261 ± 0.009[c]	2
4CP-Br	2.352 ± 0.001[d]	2.396 ± 0.001[d]	3
4BP-Br	2.352 ± 0.002[d]	2.391 ± 0.002[d]	3
4IP-Br	2.359 ± 0.001[d]	2.384 ± 0.001[d]	3
2BP-Cl	2.174 ± 0.001[d]	2.259 ± 0.001[d]	4
2IP-Cl	2.241 ± 0.001[d]	2.262 ± 0.001[d]	4
2BP-Br	2.371 ± 0.001[d]	2.407 ± 0.002[d]	4
2IP-Br	2.376 ± 0.001[d]	2.389 ± 0.001[d]	4

[a]Set 1 was excluded from this analysis since there are Cu−X bond which are proton and halogen acceptors simultaneously. [b]If there is more than one Cu−X proton acceptor, the average Cu−X distance is reported. Similarly, if there is more than one Cu−X halogen acceptor, the average is reported. [c]The error is equal to the standard deviation since there are two Cu−Cl groups which play the role of halogen bond acceptor while the other two Cu−Cl groups play the role of proton acceptor, and hence the error is large. [d]The error is extracted from the crystallographic data. Only one Cu−X group plays the role of halogen acceptor and only one plays the role of proton acceptor.

searched for iodopyridinium or substituted iodopyridium halometalate salts with C−I···X−M halogen bonding interactions with I···X distances less than the sum of van der Waals radii. Six salts were found that fulfill these parameters (Table 11). For example, type I arrangements are observed in 3IP-X structures, $(3IP)_2SnI_4$,[53] $(2A5IP)_2CuCl_4{\cdot}2H_2O$,[54] and (2A5IP)-$2CuCl_4$,[55] and type II arrangements are observed in the other structures: $(3IP)_2CoCl_4$,[14] $(3IP)_2CoBr_4$,[37] and $(3IP)_2PtCl_6{\cdot}2H_2O$ ($P\bar{1}$ and $P2/n$ phases).[56] In structures that obey the type I arrangement, the C−I···X and I···X −M angles average 169° (range = 153−175°) and 159° (range = 146−165°), while the corresponding angles in the structures that obey the type II arrangement average 174° (range 171−177°) and 105° (range = 93−117°). The similarities between the arrangements of C−Y1···Y2−C synthons and C−Y···X−M have been investigated previously.[8,37,57]

CONCLUSION

Theoretical calculations and structural analysis of compounds of the formula (nYP) CuX_4 (nYP = n-halopyrdiniumcation; Y = Cl, Br, or I; X = Cl or Br) showed that as C−Y···X−M halogen bonding interactions become stronger, the corresponding N−H···X hydrogen bond becomes weaker. To support this conclusion, the halogen and hydrogen bonding interactions were analyzed in families of isomorphous structures, allowing us to apply the separation of variables principle and hence minimize the effects of factors that arise from other intermolecular forces and the isomeric nature of halopyridinium cations (geometric factors). The above conclusion is supported by two observations:

(a) In each isomorphic structure set (Tables 8 and 10), as the organic halogen becomes heavier the halogen bond becomes stronger (while holding the halide ion constant), while the N−H···X hydrogen interactions becomes weaker. This conclusion is supported by the fact that as the organic halogen becomes heavier the N···X distance is either longer or unvaried. In contrast, Y···X distances become shorter, even though the radii of heavier halogens are larger.

(b) Theoretical calculations show that the bifurcated N−H···X hydrogen bond is stronger than the corresponding linear one. Experimentally, as the organic halogen becomes heavier, the difference between the length of the two legs of the N−H···X bifurcated hydrogen bond becomes larger (β value, Table 9) and hence closer to the linear hydrogen bond arrangement. This suggests that halogen bonding interactions reduce the role of the hydrogen bonding interactions.

Theoretical calculations have shown that hydrogen bonding interactions are always stronger than the corresponding halogn bonding interactions. This agrees with the analysis of Cu−X bond distances; the Cu−X bond distance of the proton acceptor is always longer than the corresponding distance of the halogen acceptor (Table 10).

Analysis of the C−I···X−M halogen bonding interaction angles (C−I···X and I···X −M) indicates that the arrangements of C−I···X−M synthons are similar to well studied C−Y1···Y2−C synthons.[30]

ASSOCIATED CONTENT

Supporting Information

Crystallographic information files. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

The authors declare no competing financial interest.

Table 11. Reported C−I···X−M Interactions in Cambridge Structural Database.

salt	identifier	I···X distance	C−I···X	I···X−M
$(3IP)_2CoCl_4$[14]	ALAJEC	3.213	176.77	100.74
$(3IP)_2SnI_4$[53]	BIVMID	3.560	176.65	153.18
$(2A5IP)2CuCl_4$[55a] cation 1	IGAMIM	3.379	166.19	165.37
cation 2		3.412	159.13	161.01
$(3IP)_2CoBr_4$[37] cation 1	NULQOB	3.457	175.24	93.57
cation 2		3.421	175.37	102.14
$(2A5IP)_2CuBr_4{\cdot}2H_2O$[54a] cation 1	PAXYET	3.476	175.25	156.58
cation 2		3.599	175.31	110.39
$(3IP)_2PtCl_6{\cdot}2H_2O$[56b]	YAGXUA	3.366	170.79	116.74
		3.421	173.29	104.12
$(3IP)_2PtCl_6{\cdot}2H_2O$[56c]	YAGXUA01	3.353	176.04	109.80

[a]2A5IP = 2-amino-5-iodopyridinium. [b]$P\bar{1}$ phase. [c]$P2/n$ phase.

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